A Multitask CNN-Transformer Network for Semantic Change Detection From Bitemporal Remote Sensing Images

Wei Liu, *Member, IEEE*, Ziwen Kang, Jiawei Liu, Yiyuan Lin, Yongtao Yu, *Senior Member, IEEE*, and Jonathan Li, *Fellow, IEEE*

***Abstract*— Bitemporal remote sensing (RS) semantic change detection (SCD) involves discerning and categorizing changes in the same geographical area across two RS images taken at different times. High-performance SCD approaches typically address this task using multitask networks that simultaneously handle binary change detection (BCD) and semantic segmentation (SS). Despite significant advancements in SCD research, constructing a multitask network that fully explores the correlation between BCD and SS remains challenging. To address this, we propose a novel approach called the multitask CNN-transformer network (MCTNet), tailored for SCD using bitemporal RS images. Our Siamese network simultaneously tackles SS and BCD via three subnetworks: two for SS and one for BCD. The methodology begins with a multiscale convolutional neural network (CNN) extracting local features from input images, and converting them into tokens. A Transformer module with an encoder–decoder architecture then captures long-range dependencies among these visual tokens. The extracted features are subsequently passed to multitask heads, generating predicted outputs. To ensure that the BCD results remain consistent regardless of the order of images in the input pair, we introduce spatiotemporal feature learning (SFL), enabling the acquisition of temporal-symmetric representations for BCD. Extensive experimental validation on the WHU-CD, SECOND, and HRSCD datasets demonstrates the effectiveness and efficiency of MCTNet for both SS and BCD tasks. The source code for this article will be published on GitHub in the future https://github.com/kangziwen1/MCTNet.**

***Index Terms*— Change detection (CD), multitask learning (MTL), semantic segmentation (SS), transformer.**

I. Introduction

BITEMPORAL remote sensing (RS) semantic change detection (SCD) aims to identify where changes have occurred and categorize the type of change. Unlike binary change detection (BCD), which determines simply the presence or absence of change within a target area, SCD provides more comprehensive semantic insights by presenting detailed land cover maps (LCMs) before and after change events. SCD finds application in various fields, including natural disaster assessments, environmental monitoring, agricultural surveys, and resource management. The development of high-quality satellite images taken at different times has greatly improved change detection (CD) technology. However, this has also increased demand for smarter methods to analyze these changes.

Deep learning has revolutionized CD technology in RS imagery. While existing research predominantly focuses on BCD, SCD of RS images remains relatively underexplored. From a computer vision perspective, BCD is a binary semantic segmentation (SS) task that identifies changed areas in an image. In contrast, SCD is more intricate, generating a binary change map and two temporal LCMs. Previous deep learning-based BCD models are ill-suited for SCD due to fundamental differences in their outputs and underlying assumptions. BCD models have a single output branch, while SCD models require three. Additionally, BCD models assume the invariance of the binary change map to the order of images in the input pair. In contrast, the two LCMs in SCD have a strict one-to-one correspondence with the input bitemporal images.

To address the challenges of SCD, two types of models have been proposed: single-task algorithms [1], [2] and multitask algorithms [3], [4]. Single-task SCD algorithms typically decompose the CD task into two independent stages: image SS and BCD. However, this approach relies heavily on the accuracy of the initial SS, as errors propagate to the BCD step. Moreover, the training of the SS model fails to fully account for the temporal contextual relationships between images captured in different phases. In contrast, multitask SCD algorithms use a single network to model SS and BCD jointly.

Multitask learning (MTL) offers significant benefits but poses challenges when applied to SCD. Integrating features from different tasks without causing interference or redundancy is complex. Unlike SS, high-performance BCD must remain invariant to the order of input images. Multitask heads play a critical role in guiding the learning process for each task within the MTL model, leveraging shared information across all tasks to enhance efficiency and performance. However,

Received 20 July 2024; revised 22 September 2024; accepted 18 October 2024. Date of publication 28 October 2024; date of current version 14 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62461026 and Grant 62076107 and in part by the Natural Science Foundation of Jiangxi Province under Grant 20232BAB203057 and Grant 20232BAB202004. *(Corresponding author: Yongtao Yu.)*

Wei Liu, Ziwen Kang, and Yiyuan Lin are with the School of Information and Software Engineering, East China Jiaotong University, Nanchang 330013, China.

Jiawei Liu is with the School of Information and Computer Engineering, Pingxiang University, Pingxiang 337055, China.

Yongtao Yu is with the Faculty of Computer and Software Engineering, Huaiyin Institute of Technology, Huaian 223003, China (e-mail: allennessy@hyit.edu.cn).

Jonathan Li is with the Department of Geography and Environmental Management, University of Waterloo, Waterloo, ON N2L 3G1, Canada.

Digital Object Identifier 10.1109/TGRS.2024.3486787

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

competition for shared information between tasks can lead to performance degradation. Moreover, balancing homogeneity and heterogeneity features in MTL for SCD is challenging. Homogeneity features focus on similarities between images from different times, while heterogeneity features capture their distinct characteristics. Striking the right balance between these features is crucial. Overemphasizing homogeneity may miss subtle changes, while excessive focus on heterogeneity can increase sensitivity to noise. Therefore, thoughtful design of the MTL framework is essential to mitigate these challenges and ensure optimal performance across both tasks.

In this article, we propose a multitask CNN-transformer network (MCTNet) that fully explores the correlation between BCD and SS, enabling these tasks to reinforce each other and achieve high-performance SCD. This work builds upon our previous BCD model, AMTNet [5]. Despite sharing a foundation of convolutional neural network (CNN)-Transformer structures, MCTNet and AMTNet differ significantly in their architectures. MCTNet does not employ feature exchange technology or a channel attention mechanism. Second, MCTNet designs UNet-like [6] multitask heads. Third, MCTNet integrates a spatiotemporal feature learning (SFL) module to capture BCD's symmetric temporal features.

The main contributions of this article are summarized as follows.

1) We design a MCTNet tailored to efficiently explore and exploit correlations between SS and BCD, aiming to enhance the overall performance of both tasks simultaneously.
2) We devise UNet-like [6] multitask heads for SS and BCD. These heads effectively leverage multiscale features extracted from the Transformer block, enabling the model to adapt and specialize for each specific task.
3) To learn temporal-symmetric features, we integrate an SFL module to enhance feature representation for BCD. The SFL module makes BCD invariant to the image order of the input image pair. By doing so, the model can effectively capture changes in the input data regardless of the chronological order in which they are presented.

The remaining sections of this article are structured as follows. Section II provides an overview of related work, offering insights into the existing literature. Section III elaborates on the proposed approach, offering a comprehensive discussion of the methodology. In Section IV, we present and discuss the experimental results. Finally, Section V encapsulates the conclusion and outlines potential directions for future work.

II. Related Work

A. BCD Based on Deep Learning

Contextual information is vital for CD in RS images. A series of approaches have been designed to model context and incorporate it into CNN architectures. Single-branch networks commonly aggregate features at the image level using straightforward fusion techniques like differencing, concatenation, or addition. However, these methods could not capture all relevant context information, especially for irregular or multiscale changes. Siamese network structures fuse features from two images using single-scale or multiscale fusion. Single-scale fusion [7] combines the high-level feature maps from the two temporal branches. In contrast, multiscale fusion [8], [9], [10] progressively integrates information from different levels of abstraction, building up representations from low-level to high-level features through the model hierarchy. Multiscale fusion is more effective for detecting irregular and multiscale changed objects in RS images, but it is more computationally expensive and can confuse features if not done properly. It is important to use a proper fusion method and to model contextual relationships to improve the accuracy of CD.

Transformer models have proven highly effective in computer vision, achieving SOTA results on numerous benchmarks. Their strength lies in modeling long-range dependencies, a crucial capability for many computer vision tasks. In RS, the Transformer architecture has gained increasing attention for tasks like image time-series classification [11], [12], scene classification [13], [14], hyperspectral image classification [15], [16], and CD [8], [17]. Transformers process image patches or tokens, facilitating interactions between these components to identify important regions effectively.

For CD, BIT [18] incorporates a Transformer module with a feature differencing network to capture contextual relationships. PSTNet [19] iteratively samples and optimizes features to identify changed areas gradually. Liu et al. [17] utilized CNN backbones in their MSCANet to derive hierarchical feature maps and then aggregate semantics with a Transformer. Wang et al. [8] employed multiscale Transformers to obtain multiresolution representations in their MTCNet. Changer [20] extracts multilevel feature maps from each of the two input images and then applies feature exchange between the two branches of the architecture to exchange information. This exchange integrates representations from both temporal images, improving change modeling. AMTNet [5] leverages CNNs and Transformers to extract features, uses multiscale processing to capture information at different levels, and incorporates attention to focus on relevant regions.

Despite its impressive performance in RS CD tasks, the application of the Transformer structure in CD is still in its early stages with room for improvement.

B. SCD Based on Machine Learning

SCD techniques evolved from relying on hand-engineered feature extraction to leveraging deep learning for representation learning. Early methods for SCD depended on hand-engineered features like texture and color features. These methods are limited in accurately detecting changes in complex scenes. The early intuitive algorithm [21] was from images captured at two different times to obtain land cover transition statistics. Some direct classification methods [22], [23] treat each change type as a separate class and stack multitemporal images for classification. While straightforward to implement, these techniques rely on having multitemporal labeled data covering all feasible land cover changes, which limits their applicability. To address this limitation, Bruzzone et al. [1], [2] devise a compound

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

classification approach that models the temporal relationship between two RS images through iterative semi-supervised estimation of transition probabilities. This approach does not need dedicated training data for every possible change type.

The rapid advancement of deep learning in the past decade has yielded more powerful methods for SCD. Typically, deep learning-based SCD networks comprise three subnetworks: two for SS of the individual images and one for BCD between them. For example, Daudt et al. [24] presented a dataset for SCD, and proposed four basic SCD strategies based on full CNNs, using the same encoder–decoder architecture. Liu et al. [25] used dual attention modules in their DTCDSCN architecture to improve feature discriminability through channel and spatial attention. Peng et al. [26] devised a Siamese UNet architecture that captures multiscale information and improves features via channel and spatial attention. Xia et al. [27] designed a deep Siamese postclassification fusion network (PCFN). Niu et al. [4] devised a symmetric multitask network (SMNet), using preactivated residual blocks and transformation blocks to construct the backbone. SMNet utilizes multicontent fusion modules to highlight change-related features by subtracting the extracted corresponding hierarchical features. Tian et al. [28] proposed a temporal-agnostic change region proposal network (TCRPN), based on exploring the saliency of the changed regions in single-temporal images to highlight the changed areas in the spatiotemporal features and improve the response to change. Despite these advancements, there remains room for improvement in the ability of these pure CNN Networks to capture global contextual information.

To model long-range dependencies across distant regions in RS images, several Transformer-based SCD algorithms [29], [30], [31], [32], [33] have been proposed. Shen et al. [29] introduced a global channel attention module and a multiscale feature fusion in the decoders to improve the feature representations. Ding et al. [3] introduced Bi-SRNet, a Siamese model, to integrate semantic temporal features via a deep CD unit, enhancing the network's ability to infer bitemporal semantic relation. Additionally, a loss function named Semantic Consistency Loss was proposed to improve the semantic consistency of CD results. Zheng et al. [31] introduced ChangeMask, an encoder–transformer–decoder model using Transformers for global and temporal context capture. It addresses CNN limitations with a temporally symmetric Transformer design. Cui and Jiang [32] proposed MTSCD sharing encoder layers for feature extraction and task-specific decoder branches for both tasks. Li et al. [33] presented a multitask network architecture with a decoder-focused design, emphasizing the importance of decoder features. Ding et al. [34] employed a spatiotemporal joint modeling framework to capture temporal changes, integrating spatial and temporal cues for precise detection of land cover alterations. While Transformer-based SCD algorithms have shown considerable advancements, there remains a need for further in-depth research, particularly in the realms of multiscale feature extraction and MTL.

III. Proposed Scheme

We propose a Siamese CNN-Transformer network for joint SS and BCD, as illustrated in Fig. 1. The network comprises three subnetworks: two dedicated to SS and one to BCD. The method begins by extracting local features from the input image pair using a multiscale CNN backbone, which captures visual information at various spatial resolutions. These multiscale features are then transformed into a sequence of tokens for processing by the Transformer module. The Transformer, with its encoder–decoder structure, models long-range dependencies among visual tokens. Features then pass through multitask heads to generate predictions for both tasks. For the BCD task, we enhance feature representation by incorporating SFL, which captures temporal-symmetric features and improves CD accuracy.

A. Multiscale CNN Feature Extraction

Multiscale feature extraction is crucial for capturing information at various spatial resolutions, allowing the model to extract fine-grained details and broader contextual information. This capability enables the model to discern subtle changes in land cover and land use patterns that may be imperceptible at a single spatial scale. The model enhances its discriminative power by analyzing features across multiple scales, facilitating the differentiation between land cover classes and enabling more accurate detection of semantic changes.

As depicted in Fig. 1, our MCTNet extracts multiscale feature maps from the input images by leveraging ResNet [35] as the backbone CNN. The ResNet backbone comprises a 7×7 convolutional layer followed by four residual blocks, with *Res3* and *Res4* performing $2\times$ downsampling. Noteworthy is the weight sharing between the two CNN backbones of the bitemporal Siamese network. We embed spatial positional information into these feature maps using a spatial attention module (SAM) [5]. This module enables selective focus on informative regions within the input images while suppressing irrelevant or noisy areas concurrently. Multiscale features are then fused through sampling and addition operations.

Concretely, let W and H denote the input image width and height. Given an image pair (X_1, X_2) of size $3 \times H \times W$, the CNN-based backbone extracts three feature maps at different scales from each image. Then, the multiscale feature maps pass through the SAM. Finally, for each temporal branch, the feature map $F_i^{j(2)}(\forall i \in \{1,2\}, \forall j \in \{1,2,3\})$ is merged with the other two features of the same temporal branch. This merging is done by sampling and element-wise addition operations. The output of this feature fusion process is the feature $F_i^{j(3)}$ for each input feature map $F_i^{j(2)}$.

B. Transformer Tokenization

In this stage, we convert the feature map $F_i^{j(3)}(\forall j \in \{1,2,3\})$ into a series of visual tokens to feed into the subsequent Transformer module. Following BIT [18], we utilize convolution operations to perform the tokenization, splitting the feature map $F_i^{j(3)}$ into a semantic token set $\{T_{i,k}^j\}_{k=1}^m$. Specifically, as presented in Fig. 2, we first employ a $1\times$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

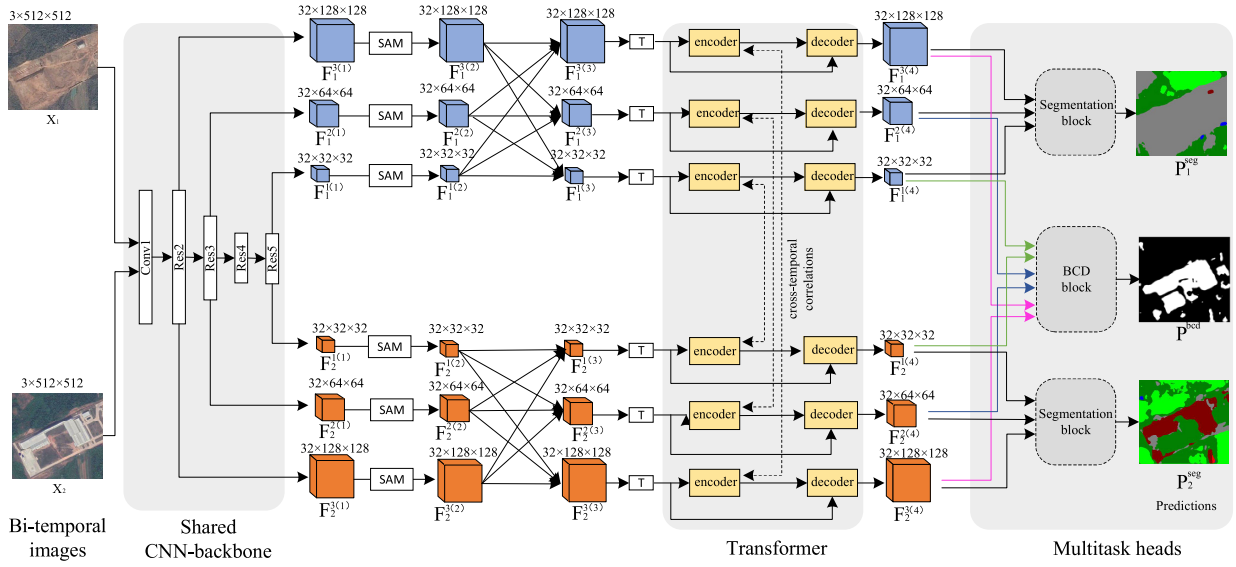


Fig. 1. Overview of our Siamese MCTNet with a shared CNN-backbone. MCTNet integrates transformers, multiscale feature fusion, and MTL efficiently. It consists of a multiscale CNN encoder, a transformer block, and multitask heads. SAM and T represent the SAM and tokenization abbreviations, respectively.

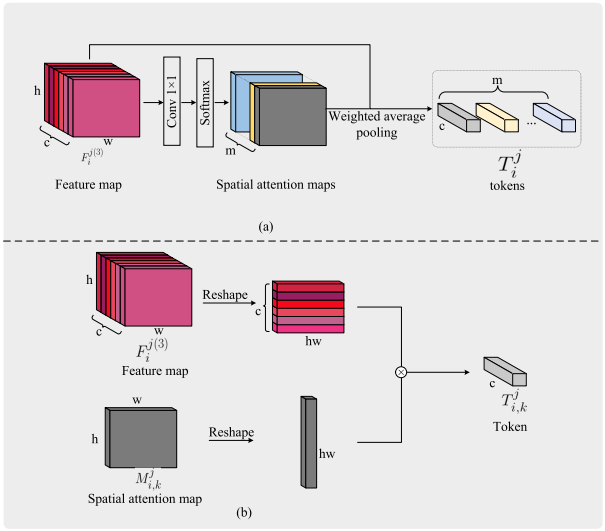


Fig. 2. (a) Tokenization process and (b) weighted average pooling illustration.

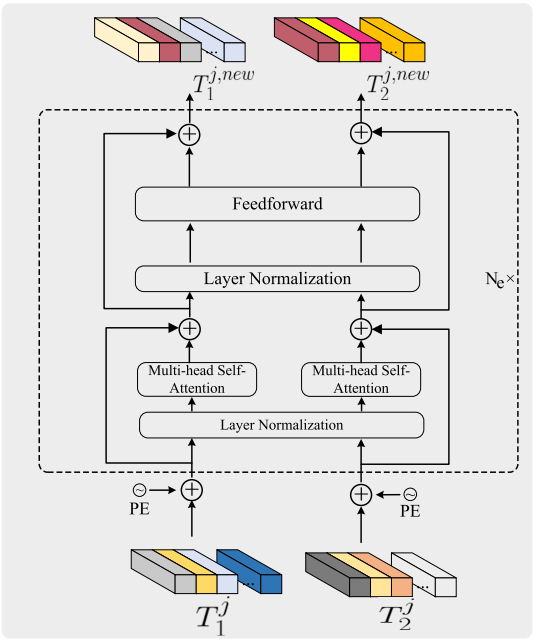


Fig. 3. Illustration of the transformer encoder block. N_e represents the number of the identical layers. PE stands for positional encoding.

1 convolutional followed by a softmax layer to $F_i^{j(3)}$ to obtain m spatial attention maps, referred to as $\{M_{i,k}^j\}_{k=1}^m$ of size $h \times w$. The attention maps represent the importance of each spatial location in $F_i^{j(3)}$. Then, we apply the weighted average pooling operation on $F_i^{j(3)}$ using the spatial attention maps. This operation computes a weighted average of the features at each spatial location, with the weights determined by the attention maps. The resulting feature maps $\{T_{i,k}^j\}_{k=1}^m$ are called semantic tokens. Reshape $F_i^{j(3)}$ and $M_{i,k}^j$ to have sizes $c \times hw$ and $hw \times 1$, respectively. The average pooling operation on $F_i^{j(3)}$ using the attention map $M_{i,k}^j$ can be denoted as

$$T_{i,k}^j\left(F_i^{j(3)}\right) = F_i^{j(3)} M_{i,k}^j. \tag{1}$$

We concatenate the semantic tokens to obtain a token sequence $T_i^j \in \mathbb{R}^{m \times c}$.

C. Transformer Encoder

This article uses transformer encoders to capture long-range dependencies via the self-attention mechanism. This approach enables our model to comprehensively understand spatial and temporal relationships within the bitemporal image pairs. As shown in Fig. 3, the Transformer encoder block takes the tokenized feature map T_i^j as input and generates an encoded output feature map $T_i^{j,\text{new}}$. The Transformer encoder has a sequence-to-sequence architecture consisting of stacked identical layers. Each layer has two key components: a multihead self-attention (MHSA) block and a feedforward network (FFN) block. Layer normalization (LN) is applied before each block.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

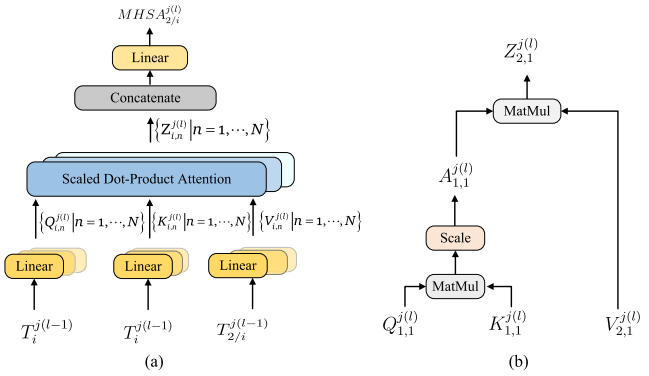


Fig. 4. Overview of the attention layer in the transformer architecture. (a) MHSA block. (b) Example of the scaled dot-production attention. For a clearer view of the details, please zoomed-in view.

The FFNs corresponding to the feature maps T_1^j and T_2^j share parameters. The MHSA blocks corresponding to the feature maps T_1^j and T_2^j do not share parameters.

Before feeding the token set $T_i^j (\forall i \in \{1, 2\})$ into the Transformer encoder, we add a learnable positional encoding $P_i^j \in \mathbb{R}^{m\times c}$ to the token set: $T_i^j = T_i^j + P_i^j$. At layer l of the Transformer encoder, the embedding $T_i^{j(l-1)}$ is passed to the MHSA, which projects it into multiple feature subspaces using N independent attention heads. The N attention heads do not share weights, and each has two main components: a transformation layer and a self-attention layer.

The Transformation layer of MHSA head n maps the input vector $T_i^{j(l-1)}$ into query ($Q_{i,n}^{j(l)} \in \mathbb{R}^{m\times d}$), key ($K_{i,n}^{j(l)} \in \mathbb{R}^{m\times d}$), and value ($V_{i,n}^{j(l)} \in \mathbb{R}^{m\times d}$). Specifically, for head n at layer l, the linear transformations can be expressed as

$$\begin{aligned} Q_{i,n}^{j(l)} &= T_i^{j(l-1)} W_{i,n}^{q,j(l)} \\ K_{i,n}^{j(l)} &= T_i^{j(l-1)} W_{i,n}^{k,j(l)} \\ V_{i,n}^{j(l)} &= T_i^{j(l-1)} W_{i,n}^{v,j(l)} \end{aligned} \tag{2}$$

where $W_{i,n}^{q,j(l)} \in \mathbb{R}^{c\times d}$, $W_{i,n}^{k,j(l)} \in \mathbb{R}^{c\times d}$, and $W_{i,n}^{v,j(l)} \in \mathbb{R}^{c\times d}$ are the linear matrices.

Given a set of keys, values, and queries, the output vector of the self-attention layer is calculated as a weighted sum of the values, using the softmax normalized attention weights. This softmax attention matrix represents the attention allocated to different token pairs. The output vectors from the N heads are concatenated and projected through a linear mapping to produce the final outputs of the self-attention layer. Specifically, for head n at layer l, the process of scale-dot attention can be formulated as

$$A_{i,n}^{j(l)} = \text{softmax}\left(\frac{Q_{i,n}^{j(l)} K_{i,n}^{j(l)^T}}{\sqrt{d}}\right) \tag{3}$$

where $(d)^{1/2}$ is a scaling factor. The softmax function normalizes the attention matrix along the row dimension. $A_{i,n}^{j(l)} \in \mathbb{R}^{m\times m}$ is the attention matrix calculating the attention weights between each pair of tokens, allowing for the allocation of attention to different parts of the input sequence.

As in Bi-SRNet [3], we project each value $V_{i,n}^{j(l)}$ using the attention matrix of the other temporal branch to capture cross-temporal correlations. Concretely, as shown in Fig. 4, the output vectors of the head n in the two temporal branches at the layer l can be expressed as

$$Z_{1,n}^{j(l)}, Z_{2,n}^{j(l)} = A_{2,n}^{j(l)} V_{1,n}^{j(l)}, A_{1,n}^{j(l)} V_{2,n}^{j(l)}. \tag{4}$$

Finally, the MHSA output at the layer l is obtained by concatenating and projecting the N head outputs

$$\text{MHA}_i^{j(l)} = \text{concat}\left(\left[Z_{i,1}^{j(l)}, Z_{i,2}^{j(l)}, \cdots, Z_{i,N}^{j(l)}\right]\right) W_i^{j(l)} \tag{5}$$

where $W_i^{j(l)} \in \mathbb{R}^{Nc\times d}$ is a projection matrix to fuse the multiple results of the heads and concat is the concatenation operation along the channel dimension.

The MHSA output is then passed to the FFN block, which comprises two linear layers with a GELU activation [36] in between. The GELU activation helps introduce nonlinearity to learn complex input–output relationships.

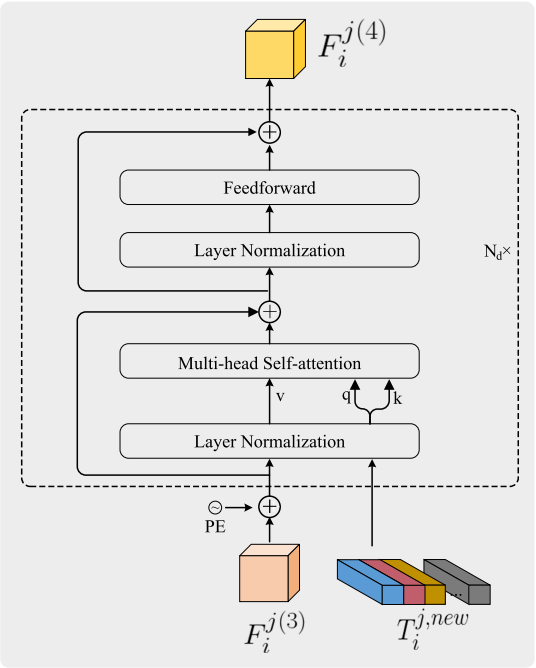


Fig. 5. Illustration of the transformer decoder process. The feature $F_i^{j(3)}$ and 3-D token embedding $T_i^{j,\text{new}}$ are input at the same time, resulting in the output feature $F_i^{j(4)}$. N_d represents the number of the identical layers.

D. Transformer Decoder

The decoder block uses the encoder's output as its input and reconstructs pixel-level features in pixel space. As shown in Fig. 5, the token decoder takes the convolutional feature $F_i^{j(3)}$ and the output $T_i^{j,\text{new}}$ from the token encoder as input. Before being fed to the decoder, a trainable positional encoding is added to $F_i^{j(3)}$. Like the encoder, the decoder also adopts a sequence-to-sequence architecture that consists of a stack of identical layers. However, the decoder takes a different approach to mapping the input features. Specifically, the decoder maps $F_i^{j(3)}$ into the query and maps $T_i^{j,\text{new}}$ into the key and the value. After the feature $F_i^{j(3)}$ is processed by the Transformer module, feature $F_i^{j(4)}$ will be obtained.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

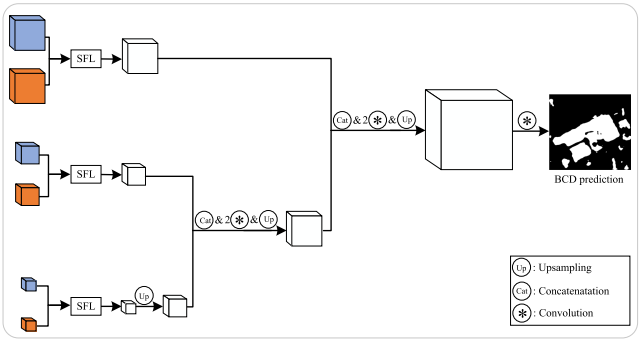


Fig. 6. Illustration of the BCD head.

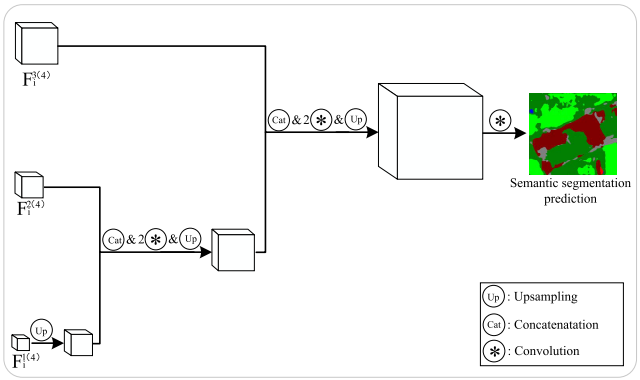


Fig. 7. Illustration of the SS head.

E. Multitask Heads and the Total Loss

1) Multitask Heads: The multitask heads enable the model to leverage shared representations across different tasks, improving the performance of both tasks. SS and BCD are both pixel-wise classification tasks. We use the same network architecture for both tasks to simplify the model, with only a few modifications. The two SS heads share weights, while the BCD head has independent weights. As shown in Figs. 6 and 7, the architecture employed is a UNet-like [6] network containing three $2\times$ upsampling blocks. Each upsampling block uses nearest-neighbor upsampling, and two 3×3 convolutional layers are applied before two of the upsampling blocks. Besides the SFL module, the main difference between the SS heads and the BCD head is in the number of output channels. The BCD head has two channels for changed and unchanged areas, while the SS heads have channels equal to the number of predefined semantic classes.

2) SFL: SFL has been shown effective for BCD tasks [31], [37]. Inspired by prior research, we incorporate an SFL module into our multiscale network to facilitate temporal-symmetric feature acquisition for the BCD task. This SFL module imparts resilience to the order of input images. Our model discerns pertinent features and yields consistent BCD outcomes whether the images are arranged chronologically or in reverse order. This immunity to image sequence is pivotal for ensuring the precision and dependability of BCD, safeguarding against biases that might arise from the arrangement of input images.

As shown in Fig. 8, the SFL module works as follows. First, we concatenate $F_1^{j(4)} \in \mathbb{R}^{c\times h\times w}$ and $F_2^{j(4)} \in \mathbb{R}^{c\times h\times w}$ along the

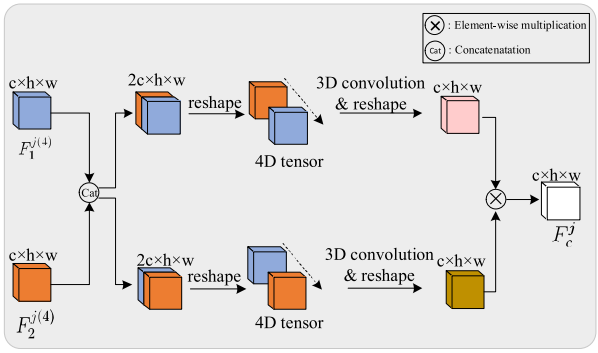


Fig. 8. BCD task's SFL module illustration.

channel dimension to get two new features $F_{1\to 2}^{j(4)}$ and $F_{2\to 1}^{j(4)}$ of size $2c \times h \times w$. Formally,

$$F_{1\to 2}^{j(4)} = \text{concat}\left(F_1^{j(4)}, F_2^{j(4)}\right)$$
$$F_{2\to 1}^{j(4)} = \text{concat}\left(F_2^{j(4)}, F_1^{j(4)}\right). \tag{6}$$

$F_{1\to 2}^{j(4)}$ and $F_{2\to 1}^{j(4)}$ are concatenated in the reverse order.

Next, we reshape the size of the two concatenated feature maps from $2c \times h \times w$ to $2 \times c \times h \times w$

$$F_{1\to 2}^{j(4)}, F_{2\to 1}^{j(4)} = \text{reshape}\left(F_{1\to 2}^{j(4)}\right), \text{reshape}\left(F_{2\to 1}^{j(4)}\right). \tag{7}$$

This gives us two 4-D feature maps, where the first and second dimensions of the two features correspond to the temporal phase and feature channel, respectively, and the third and fourth dimensions correspond to the spatial dimensions.

Finally, we apply a 3-D convolution with a kernel size of $k \times k \times k$ to each of the two 4-D features. We then fuse the two spatiotemporal features using element-wise multiplication to obtain the final fused feature

$$F_c^j = \text{Conv3d}\left(F_{1\to 2}^{j(4)}\right) \otimes \text{Conv3d}\left(F_{2\to 1}^{j(4)}\right) \tag{8}$$

where $\otimes$ is element-wise multiplication and Conv3d is the 3-D convolution. This operation is symmetric, implying that the final fused feature F_c^j remains unaffected by the sequence in which the two feature maps, $F_{1\to 2}^{j(4)}$ and $F_{2\to 1}^{j(4)}$, are concatenated.

3) Total Loss: The overall loss of our MCTNet consists of three losses: two SS losses and one BCD loss. We use multiclass cross-entropy (CE) loss to model the SS losses and binary CE loss to model the BCD loss. Specifically, let $\mathcal{L}^{\text{seg1}}$ and $\mathcal{L}^{\text{seg2}}$ denote SS losses with respect to temporal T_1 and temporal T_2, respectively. The two SS losses can be expressed as

$$\mathcal{L}^{\text{seg1}} = -\frac{1}{N_p}\sum_{i=1}^{N_p} y_i^{\text{seg1}} \log\left(p_i^{\text{seg1}}\right)$$
$$\mathcal{L}^{\text{seg2}} = -\frac{1}{N_p}\sum_{i=1}^{N_p} y_i^{\text{seg2}} \log\left(p_i^{\text{seg2}}\right) \tag{9}$$

where N_p is the number of pixels in the training set. y_i^{seg1} and y_i^{seg1} are the ground truth (GT) labels of the ith pixel pair for the SS task. p_i^{seg1} denotes the predicted probability of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

branch corresponding to temporal T_1 for the ith pixel with respect to the class y_i^{seg1}, likewise p_i^{seg2}.

The BCD loss $\mathcal{L}_{\text{bcd}}$ between the BCD predicted results and the GT can be formulated as

$$\mathcal{L}^{\text{bcd}} = -\frac{1}{N_p}\sum_{i=1}^{N_p}\left(y_i^{\text{bcd}}\log\left(p_i^{\text{bcd}}\right) - \left(1-y_i^{\text{bcd}}\right)\log\left(1-p_i^{\text{bcd}}\right)\right) \tag{10}$$

where y_i^{bcd} and p_i^{bcd} represent the GT and the predicted result of the ith pixel pair, respectively.

The total loss of the MCTNet is the sum of the SS losses and the BCD loss

$$\mathcal{L} = \lambda\left(\mathcal{L}^{\text{seg1}} + \mathcal{L}^{\text{seg2}}\right) + \mathcal{L}^{\text{bcd}} \tag{11}$$

where λ represents the weights of the SS losses. In practice, the value of λ can be set by cross-validation.

IV. Experiments

A. Settings

1) Datasets: We performed comprehensive experiments on three SCD benchmark datasets: WHU-CD [38], SECOND [39], and HRSCD [24].

a) WHU-CD: It contains two 32 507 × 15 354-pixel aerial images of Christchurch, New Zealand captured in 2012 and 2016 at a 0.2-m/pixel spatial resolution. The dataset includes labels for buildings and building changes between the 2012 and 2016 images. This work divides both images into nonoverlapping 256 × 256-pixel patches for model training and evaluation. The dataset contains 4461 training patches, 1487 validation patches, and 1486 test patches.

b) SECOND: It contains 4462 pairs of RS images from various sensors, with publicly available annotations for 2968 image pairs. Each 512 × 512-pixel image has a spatial resolution of 0.3–5 m and covers areas in Hangzhou, Chengdu, and Shanghai. It provides pixel-level annotations for six land cover categories, including *nonvegetated ground*, *trees*, *low vegetation*, *water*, *buildings*, and *playgrounds*. It includes labels for 37 potential land cover change types, including 36 interclass changes and unchanged areas. The dataset is split into a training set of 2375 image pairs and a test set of 593 pairs, a 4:1 ratio of training to test data.

c) HRSCD: It comprises 291 pairs of RGB aerial images with a spatial resolution of 0.5 m and a size of 10 000 × 10 000 pixels. Each image pair is accompanied by corresponding land cover annotations, detailing six distinct categories: *No information*, *artificial surface*, *agricultural area*, *forest*, *wetland*, and *water*. These images were meticulously sourced from urban and rural locales in Rennes and Caen, France. We employed a nonoverlapping cropping approach on the original images. Images were cropped only when the proportion of the changed area exceeds 10%. We derived 2373 pairs of 512 × 512-pixel samples for model training and evaluation through this process. Pixels labeled as *No information* were excluded from the training dataset, ensuring CD algorithms are assessed solely on the remaining five categories. The dataset was divided into 1900 training image pairs and 473 testing image pairs, maintaining a 4:1 training-to-test ratio.

2) Metrics: We conducted a comparative analysis of our MCTNet with several benchmark algorithms on both the BCD and SCD tasks. Following the setting of AMTNet [5], we report results using four common evaluation metrics for BCD: precision, recall, intersection over union (IoU), and $F1$ score. For SS, we report performance metrics for individual categories using the same metrics. Additionally, we compare the average performance across all categories. To ensure a comprehensive evaluation of SS performance, we include overall accuracy (OA), average precision (AP), and the Kappa coefficient in our analysis.

B. Comparison Algorithms

We compare our method against recent SOTA CD algorithms including CNN-based and Transformer-based approaches. The pure CNN-based comparison algorithms consist of DTCDSCN [25], DSIFN/IFN [40], MFPNet [41], and SNUNet [42]. The Transformer-based comparison algorithms consist of BIT [18], MSCANet [17], Bi-SRNet [3], ChangeFormer [43], and AMTNet [5]. Among the comparison algorithms, DTCDSCN and Bi-SRNet are multitask networks, while the others are single-task networks. All comparison algorithms have publicly available implementations.

Implementation Details: We implemented MCTNet and baseline models in PyTorch, using an RTX 3090 GPU for training and evaluation. We standardized the input image sizes: 256 × 256 pixels for WHU-CD and 512 × 512 pixels for SECOND and HRSCD, with a batch size of 8. We optimized using the AdamW optimizer [44] with an initial learning rate of 1e-4 and a weight decay of 0.01. MCTNet used standard CE loss for both the BCD and SS tasks when compared with the baselines for a fair comparison. The Transformer module contained four encoders and decoders. For the SFL stage, we set the 3-D convolution kernel size to 3 × 3 × 3. Training included random rotations and flips for data augmentation. The token length l was 4. Models underwent training for 100 epochs. Training times were approximately 2.6 h for WHU-CD, 4.8 h for SECOND, and 3.8 h for HRSCD with a ResNet-50 backbone. Testing did not use data augmentation. The loss weight λ was set to 0.5 for WHU-CD, 0.01 for SECOND, and 0.1 for HRSCD through cross-validations.

C. Comparison Results

1) Comparison Results on BCD:

a) WHU-CD: As presented in Table I, our MCTNet attains new SOTA results on this dataset. Specifically, MCTNet obtains the highest recall at 95.24%, the best $F1$ score at 94.35%, and the highest IoU at 89.30%. It also achieves the second-highest precision at 93.47%. The next top-performing model is Bi-SRNet. It achieves strong results with a precision of 92.58%, recall of 94.39%, $F1$ score of 93.48%, and IoU of 87.76%. Though slightly below MCTNet, Bi-SRNet still demonstrates excellent performance overall, coming in second across key evaluation metrics. MCTNet improves precision by 0.89%, recall by 0.85%, $F1$ score by 0.87%, and IoU by 1.54% relative to Bi-SRNet. The single-task Siamese CNN-Transformer network AMTNet achieves the third-best overall

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON RESULTS FOR THE BCD TASK ON WHU-CD (%)

Approach	WHU-CD			
	Precision	Recall	F1	IoU
DTCDSCN* [25]	83.78	**92.99**	88.14	78.79
DSIFN/IFN [40]	**95.88**	86.50	90.95	83.40
SNUNet* [42]	86.33	92.31	89.22	80.54
MFPNet [41]	83.72	92.23	87.77	78.20
BIT [18]	74.40	88.56	80.87	67.88
ChangeFormer [43]	87.23	80.80	83.89	72.26
Bi-SRNet [3]	**92.58**	**94.39**	**93.48**	**87.76**
MSCANet [17]	90.30	91.09	90.69	82.97
AMTNet [5]	92.32	90.57	**91.44**	**84.23**
MCTNet (Ours)	**93.47**	**95.24**	**94.35**	**89.30**

Results marked with "*" are unofficial re-implementations of the prior methods. Color convention: **best**, **2nd-best**, and **3rd-best**.

TABLE II
COMPARISON RESULTS FOR THE BCD TASK ON SECOND (%)

Approach	SECOND			
	Precision	Recall	F1	IoU
DTCDSCN* [25]	**73.83**	61.09	66.86	50.21
DSIFN/IFN [40]	71.02	63.08	66.82	50.17
SNUNet* [42]	72.74	62.00	66.94	50.31
MFPNet [41]	**75.71**	61.66	**67.97**	**51.48**
BIT [18]	73.73	62.07	67.40	50.83
ChangeFormer [43]	70.04	50.93	58.97	41.82
Bi-SRNet [3]	71.98	**65.07**	**68.35**	**51.92**
MSCANet [17]	68.44	**63.21**	65.72	48.95
AMTNet [5]	71.53	62.96	66.97	50.34
MCTNet (Ours)	**73.80**	**66.75**	**70.10**	**53.96**

TABLE III
COMPARISON RESULTS FOR THE BCD TASK ON HRSCD (%)

Approach	HRSCD			
	Precision	Recall	F1	IoU
DTCDSCN* [25]	72.34	65.39	68.69	52.31
DSIFN/IFN [40]	71.73	71.58	71.66	55.83
SNUNet* [42]	71.97	71.56	71.77	55.97
MFPNet [41]	74.63	68.98	71.69	55.88
BIT [18]	**75.86**	64.08	69.47	53.22
ChangeFormer [43]	74.51	63.34	68.47	52.06
Bi-SRNet [3]	**75.53**	71.07	**73.23**	**57.76**
MSCANet [17]	72.29	**72.11**	72.20	56.50
AMTNet [5]	72.75	**74.21**	**73.47**	**58.07**
MCTNet (Ours)	**77.43**	**72.40**	**74.83**	**59.78**

performance. Compared with AMTNet, the improvement in the performance of MCTNet and Bi-SRNet is likely due to the multitask architecture. Improving recall [capturing more true positives (TPs)] may decrease precision [more false positives (FPs)], and vice versa. As for DSIFN/IFN, it has the highest precision but not a high recall rate. As shown in Table I, the performance of the comparison algorithms is already very high, and there is not much room for performance improvement.

b) SECOND: As presented in Table II, MCTNet obtains the best overall results for the BCD task, with a recall of 66.75%, $F1$ score of 70.10%, and IoU of 53.96%. The second-best overall performer is Bi-SRNet. It achieves strong results with a precision of 65.07%, recall of 68.35%, $F1$ score of 66.67%, and IoU of 51.92%. MCTNet improves precision by 1.82%, recall by 1.68%, $F1$ score by 1.75%, and IoU by 2.04% relative to Bi-SRNet. The pure CNN model MFPNet achieves the third-best results overall on this dataset. MFPNet's strong performance can be attributed to its sophisticated architecture, which incorporates multiscale features and attention modules.

c) HRSCD: As depicted in Table III, our MCTNet demonstrates superior performance for the BCD task on HRSCD, achieving the highest overall results. It achieves a precision of 77.43%, recall of 72.40%, $F1$ score of 74.83%, and IoU of 59.78%. AMTNet, a CNN-transformer network, achieves the second-best overall results on this dataset, with a precision of 72.75%, recall of 74.21%, $F1$ score of 73.47%, and IoU of 58.07%. Its robust performance can be credited to its complex architecture, which integrates ConvNets, transformers, multiscale, and attention mechanisms to capture spatial dependencies effectively. Compared to AMTNet, MCTNet enhances precision by 4.68%, $F1$ score by 1.36%, and IoU by 1.71%. The third-best performer is Bi-SRNet, which also exhibits strong results with a precision of 75.53%, recall of 71.07%, $F1$ score of 73.23%, and IoU of 57.76%.

2) Performance Comparison on SS:

a) WHU-CD: This dataset contains only two semantic classes: *background* and *building*. As shown in Table IV, our MTCNet achieves the best performance for the SS task across both categories in this dataset. Bi-SRNet and DTCDSCN achieve the second- and third-best overall performance, respectively. The performance gap between MTCNet and Bi-SRNet is small on this dataset. The pure CNN-based DTCDSCN model achieves significantly lower $F1$ and IoU scores than the two CNN-Transformer hybrid networks. Our MTCNet obtains

TABLE IV
SS COMPARISON RESULTS ON WHU-CD (%)

Metric	Approach	Class	
		Background	Building
Precision	DTCDSCN	98.12	81.88
	Bi-SRNet	99.06	96.09
	MTCNet	**99.11**	**96.51**
Recall	DTCDSCN	96.29	90.08
	Bi-SRNet	99.17	95.58
	MTCNet	**99.26**	**95.83**
F1	DTCDSCN	97.19	85.78
	Bi-SRNet	99.11	95.83
	MTCNet	**99.19**	**96.17**
IoU	DTCDSCN	94.54	75.11
	Bi-SRNet	98.24	92.00
	MTCNet	**98.38**	**92.61**

TABLE V
SS COMPARISON RESULTS (%)

Dataset	Approach	mIoU	AP	OA	Kappa
WHU-CD	DTCDSCN	84.82	90.00	95.31	82.98
	Bi-SRNet	95.12	97.57	98.54	94.95
	MTCNet	**95.5**	**97.81**	**98.66**	**95.35**
SECOND	DTCDSCN	46.77	67.66	74.78	64.47
	Bi-SRNet	48.29	69.07	74.44	64.24
	MTCNet	**54.84**	**73.18**	**77.65**	**68.78**
HRSCD	DTCDSCN	48.21	60.83	81.72	66.11
	Bi-SRNet	40.52	62.57	62.30	31.09
	MTCNet	**62.68**	**73.56**	**87.07**	**76.19**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON RESULTS OF THE SS TASK PERFORMED ON SECOND (%)

Metric	Methond	Class					
		Water	Ground	Low vegetation	Tree	Building	Playground
Precision	DTCDSCN	44.74	74.50	**70.37**	60.98	82.24	71.33
	Bi-SRNet	53.18	75.15	68.13	55.94	84.78	77.22
	MTCNet	**64.70**	**80.16**	68.23	**62.96**	**87.04**	**75.99**
Recall	DTCDSCN	20.67	**79.28**	61.28	58.48	87.83	43.02
	Bi-SRNet	19.55	78.43	61.73	**65.94**	85.06	52.82
	MTCNet	**46.00**	78.35	**70.88**	62.12	**88.51**	**58.93**
F1	DTCDSCN	28.27	76.81	65.51	59.70	84.94	53.67
	Bi-SRNet	28.59	76.76	64.77	60.53	84.92	62.73
	MTCNet	**53.77**	**79.24**	**69.53**	**62.54**	**87.77**	**66.38**
IoU	DTCDSCN	16.47	62.36	48.71	42.26	73.83	36.68
	Bi-SRNet	16.68	62.28	47.90	43.40	73.79	45.70
	MTCNet	**36.77**	**65.62**	**53.29**	**45.49**	**78.20**	**49.68**

TABLE VII
COMPARISON RESULTS OF THE SS TASK PERFORMED ON HRSCD (%)

Metric	Methond	Class				
		Artificial surface	Agricultural area	Forest	Wetland	Water
Precision	DTCDSCN	81.35	84.69	58.31	12.86	66.96
	Bi-SRNet	51.72	70.52	63.02	37.71	**89.89**
	MTCNet	**86.91**	**89.40**	**63.73**	**44.86**	82.91
Recall	DTCDSCN	78.42	86.18	65.12	18.92	61.91
	Bi-SRNet	64.31	62.49	38.78	33.75	63.92
	MTCNet	**83.93**	**89.86**	**81.70**	**54.19**	**80.96**
F1	DTCDSCN	79.86	85.43	61.52	15.31	64.22
	Bi-SRNet	57.33	66.26	48.01	35.62	74.71
	MTCNet	**85.39**	**89.63**	**71.60**	**49.08**	**81.92**
IoU	DTCDSCN	66.47	74.56	44.33	8.30	47.29
	Bi-SRNet	40.18	49.54	31.59	21.67	59.63
	MTCNet	**74.51**	**81.21**	**55.77**	**32.52**	**69.38**

strong results with a precision of 96.51%, recall of 95.83%, $F1$ score of 96.17%, and IoU of 92.61%. The IoU of Bi-SRNet and MTCNet on this dataset already exceeds 90%, indicating that performance is near saturation. Furthermore, Table V highlights that our MTCNet achieves the most superior SS results among all SCD methods on the WHU-CD dataset, as evidenced by metrics including mIoU, AP, OA, and Kappa.

b) SECOND: Compared to WHU-CD, the SECOND dataset poses a heightened challenge for SS due to its six distinct categories. Notably, it is highly imbalanced, with over 80% of the samples falling into the *ground*, *low vegetation*, and *building* classes. Table VI compares the three SCD approaches for each class. Our MCTNet demonstrates superior overall performance in terms of precision, recall, $F1$ score, and IoU. Particularly for the *water* class, MCTNet outperforms Bi-SRNet and DTCDSCN. All three SCD models exhibit strong recognition capabilities for *ground*, *low vegetation*, and *buildings* across the four evaluation metrics in Table VI. From Fig. 9, it is evident that *water* and *trees* are sometimes misclassified as *low vegetation*, and *playground* as *building*. Although MCTNet achieves state-of-the-art performance on the SECOND dataset, further enhancements are needed, particularly in terms of recall for the rare classes such as *water*, *low vegetation*, and *trees*.

c) HRSCD: As presented in Table VII, except for the precision for *water*, our MCTNet outperforms the other counterparts on HRSCD across all four evaluation metrics for all categories. Especially category *wetlands*, the performance of our algorithm is significantly better than the other two algorithms. As observed from Fig. 9, Bi-SRNet tends to misclassify *forests* as *artificial surfaces* or *agricultural areas*, and it also tends to misidentify wetlands as *artificial surfaces*. DTCDSCN tends to misclassify *wetlands* as *agricultural areas*. Table V highlights that our MTCNet achieves the most superior SS results among all SCD methods on the HRSCD dataset, as evidenced by all four evaluation metrics, including mIoU, AP, OA, and Kappa.

3) Visualization Comparison: Fig. 10 visually compares BCD results on three SCD benchmarks. In Fig. 10, a better result has fewer red and green pixels overall. BCD aims to maximize the number of white and black pixels, reflecting more accurate segmentation of the true classes, while minimizing the number of red and green pixels representing incorrect predictions. Fig. 10 shows that our MCTNet outperforms the comparison algorithms. Most comparative algorithms struggle to precisely identify changes occurring in irregularly shaped buildings across various sizes. However, our MCTNet can still clearly delineate building boundaries.

TABLE VIII
EFFICIENCY COMPARISON ANALYSIS WITH DIFFERENT INPUT SIZES

Method	512x512			256x256		
	FLOPs(G)	Parms(M)	FPS	FLOPs(G)	Parms(M)	FPS
DTCDSCN [25]	52.83	31.26	2.31	13.22	31.26	6.01
BiT [18]	42.37	3.49	6.40	8.75	3.49	28.51
SNUNet [42]	219.33	12.03	0.90	54.83	12.03	4.75
MFPNet [41]	514.93	85.97	0.62	128.85	85.97	2.78
ChangeFormer [43]	230.25	32.03	0.49	202.79	32.03	1.95
DSIFN/IFN [40]	329.06	50.44	1.06	82.26	50.44	4.16
MSCANet [17]	59.08	16.42	4.10	14.80	16.42	15.57
Bi-SRNet [3]	190.30	23.38	1.65	47.54	23.38	7.67
AMTNet [5]	86.23	24.67	2.61	21.56	24.67	9.02
MCTNet(Ours)	75.48	24.90	2.86	18.87	24.90	8.67

Fig. 11 visually assesses the SCD task using randomly selected samples. Our proposed MCTNet and Bi-SRNet perform quite well on WHU-CD, exhibiting an equivalently high level of competence. When confronted with SECOND and HRSCD, MCTNet surpasses DTCDSCN and Bi-SRNet, precisely pinpointing and categorizing change types. Consistent with the observation from Tables V–VII, our MCTNet model significantly outperforms the other two SCD methods in water segmentation.

4) Model Efficiency: Three key metrics were assessed to evaluate model efficiency: floating point operations (FLOPs), parameters (Params), and frames per second (FPS). Table VIII summarizes these metrics for inputs of 256×256 pixels and 512×512 pixels. MFPNet tops FLOPs and params due to its complex multiscale design. Of the transformer-based networks, ChangeFormer exhibits the highest computational complexity and parameter count resulting from its hierarchical Transformer encoder. Our MCTNet has 75.48 G FLOPs and 24.90 M Params, given 512×512-pixel input images. Reducing the input size to 256×256 pixels decreases these to 18.87 G FLOPs and 24.90 M Params. MCTNet outperforms the multitask models DTCDSCN and Bi-SRNet in terms of processing speed. Table VIII indicates that our MCTNet achieves SOTA SCD performance while maintaining low FLOPs and Params and without significant compromise in processing speed.

5) Discussion: The experimental results reveal three key observations.

1) Transformer-based SCD networks surpass their BCD counterparts for the CD task across all three CD datasets. The integration of land-cover categorical information enhances performance. By utilizing MTL, these SCD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

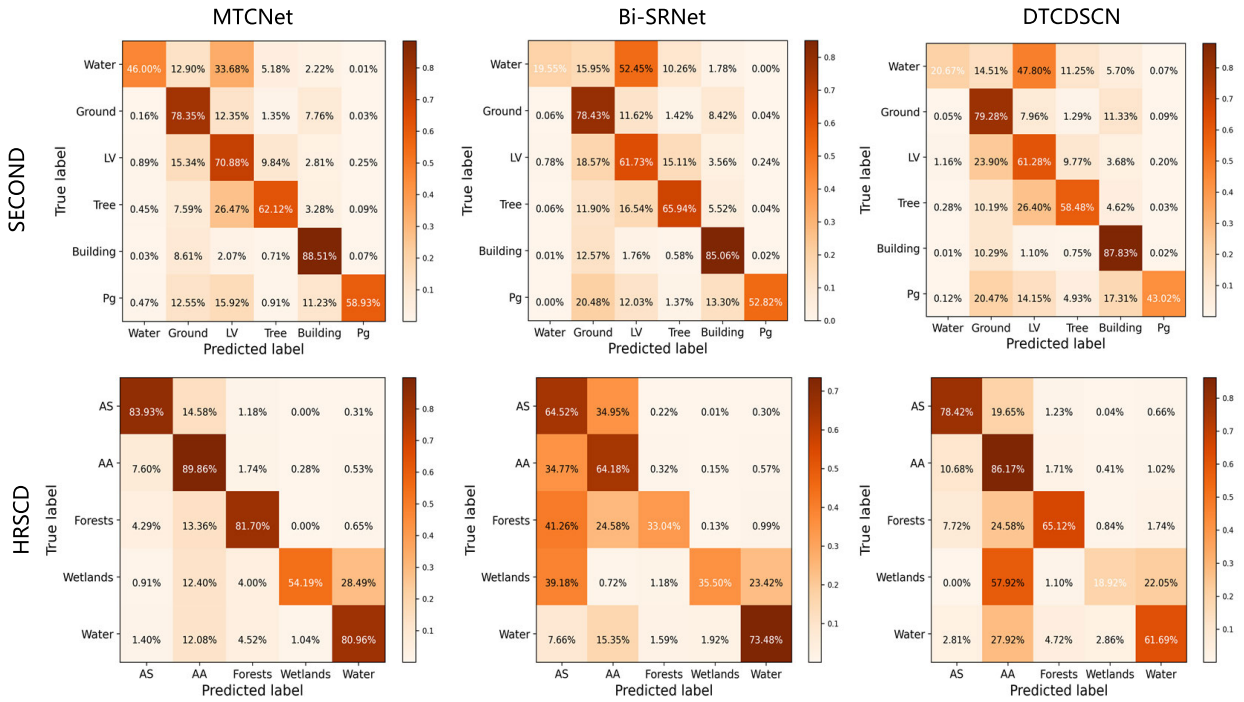


Fig. 9. Confusion matrices for SS tasks using three SCD algorithms on SECOND and HRSCD datasets. AA, AS, LV, and Pg are abbreviations for *artificial surface*, *agricultural area*, *low vegetation*, and *playground*, respectively.

TABLE IX

IMPACT OF THE NUMBER OF TRANSFORMER ENCODERS AND DECODERS FOR BCD (%)

Enc.	Dec.	WHU-CD		SECOND		HRSCD	
		F1	IoU	F1	IoU	F1	IoU
1	1	93.49	87.77	69.77	53.57	73.88	58.58
2	2	93.56	87.89	69.10	52.79	74.07	58.81
3	3	93.65	88.06	69.30	53.03	74.70	59.61
4	4	**94.35**	**89.30**	**70.10**	**53.96**	**74.83**	**59.78**
5	5	93.94	88.57	69.37	53.10	74.30	59.11

"Enc." and "Dec." are abbreviations for "encoder" and "decoder," respectively.

TABLE X

IMPACT OF THE NUMBER OF TRANSFORMER ENCODERS AND DECODERS FOR THE SS TASK (%)

Enc.	Dec.	WHU-CD		SECOND		HRSCD	
		mIoU	Kappa	mIoU	Kappa	mIoU	Kappa
1	1	95.10	94.93	55.97	69.13	58.23	72.28
2	2	95.12	94.95	54.70	68.40	60.89	75.22
3	3	95.45	95.30	53.55	69.06	62.83	75.50
4	4	**95.50**	**95.35**	54.84	68.78	**62.68**	**76.19**
5	5	95.28	95.12	**56.18**	**69.09**	62.01	75.39

networks exploit correlations between tasks to boost overall performance.

2) Our method showcases superior effectiveness compared to SOTA SCD networks DTCDSCN and Bi-SRNet.
3) Our SCD network exhibits robust performance and high efficiency.

D. Impact of Different Model Settings

In this section, we investigate the impact of various model configurations including the number of encoders and decoders, convolution kernel size in the SFL, the SS loss weight λ, and types of SS losses.

TABLE XI

IMPACT OF THE CONVOLUTION KERNEL SIZE IN SFL FOR BCD (%)

Size		WHU-CD		SECOND		HRSCD	
Kernel	Padding	F1	IoU	F1	IoU	F1	IoU
3	1	**94.35**	**89.30**	**70.10**	**53.96**	**74.83**	**59.78**
5	2	93.70	88.15	69.60	53.37	74.56	59.43
7	3	93.80	88.32	69.33	53.06	74.39	59.23

TABLE XII

IMPACT OF THE CONVOLUTION KERNEL SIZE IN SFL FOR SS (%)

Size		WHU-CD		SECOND		HRSCD	
Kernel	Padding	mIoU	Kappa	mIoU	Kappa	mIoU	Kappa
3	1	**95.50**	**95.36**	54.84	68.78	**62.68**	**76.19**
5	2	95.27	95.10	**56.10**	**69.24**	59.26	75.48
7	3	95.39	95.24	54.78	68.70	58.57	74.92

1) Model Performance Using Different Encoder-Decoder Configurations: Tables IX and X compare the performance of MTCNet using different numbers of encoders and decoders. Our experiments demonstrate that just a single encoder and decoder pair yields robust outcomes for both tasks. As we increase the number of encoders and decoders, the performance improves, generally peaking at four encoders and four decoders for both tasks. These experimental results indicate that the setting of four encoders and four decoders offers an optimal balance between model effectiveness and computational efficiency. Going beyond four encoders and decoders offers diminishing returns and is likely not worthwhile given the increased computational requirements.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

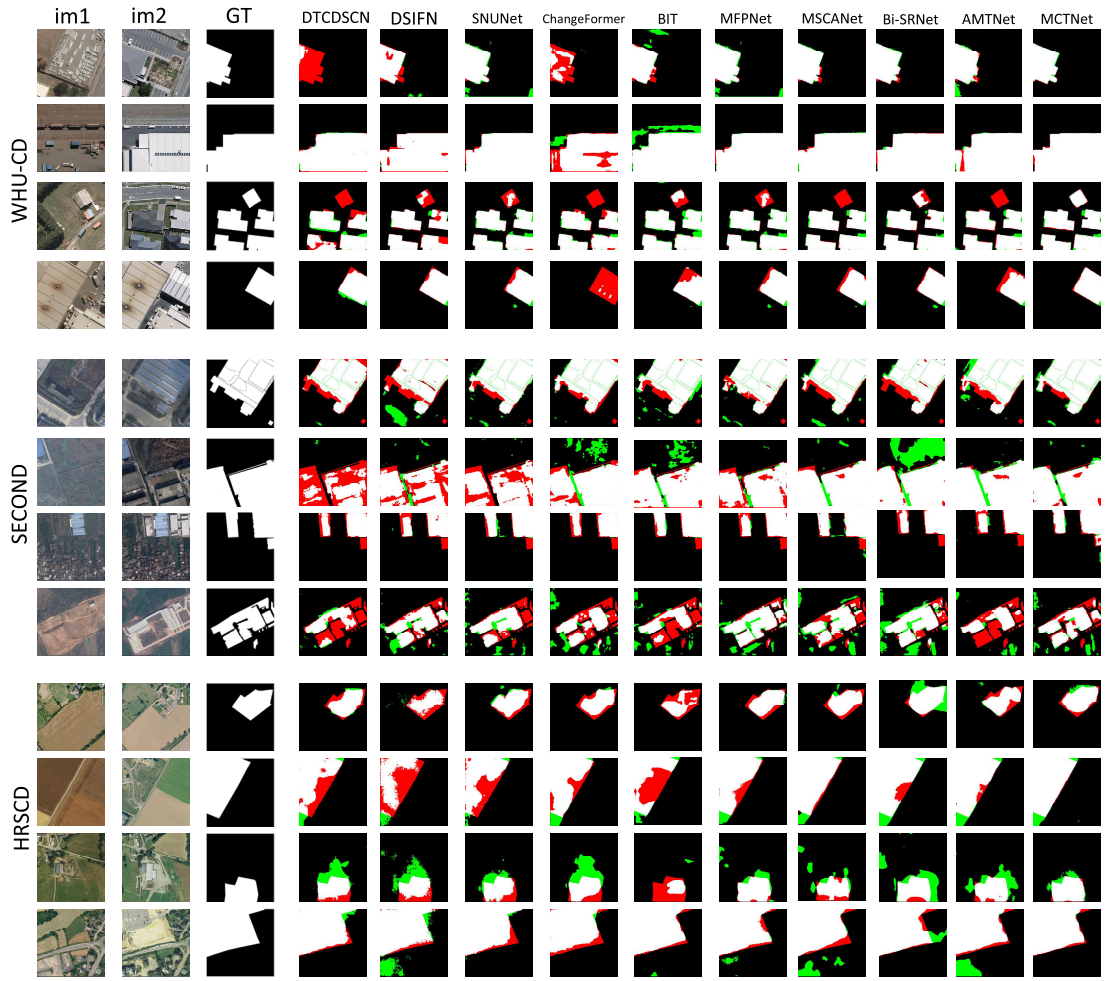


Fig. 10. Visual comparison of BCD results. White pixels indicate TP, while black pixels denote true negatives (TNs). Green pixels represent FP, and false negatives (FNs) are shown in red.

TABLE XIII
IMPACT OF SS ON BCD WITH VARYING λ (%)

λ	WHU-CD		SECOND		HRSCD	
	F1	IoU	F1	IoU	F1	IoU
0	91.78	84.81	69.94	53.78	74.29	59.10
0.01	92.23	85.78	**70.10**	**53.96**	74.71	59.63
0.1	93.08	87.06	70.06	53.91	**74.83**	**59.78**
0.3	92.97	86.86	69.83	53.65	74.67	59.58
0.5	**94.35**	**89.30**	69.33	53.05	74.35	59.17
1	93.55	87.89	68.58	52.18	74.17	58.95
10	91.88	84.97	66.99	50.36	73.46	58.05

TABLE XIV
BCD RESULTS USING DIFFERENT SS LOSSES (%)

Dataset	Loss	Precision	Recall	F1	IoU
WHU-CD	CE	**93.47**	95.24	**94.35**	**89.30**
	WCE	92.94	**95.48**	94.19	89.02
SECOND	CE	73.80	66.75	70.10	53.96
	WCE	**73.89**	**67.16**	**70.37**	**54.28**
HRSCD	CE	**77.43**	72.40	**74.83**	**59.78**
	WCE	75.18	**73.61**	74.38	59.22

2) Impact of the Convolution Kernel Size in the SFL: We also assessed the impact of convolution kernel sizes within the SFL module, ranging from 3×3 to 7×7, as detailed in Tables XI and XII. The model performs well across all tested configurations for both tasks, but optimal results are achieved with a kernel size of 3 and a padding size of 1.

3) Impact of SS on BCD: Table XIII illustrates the performance of BCD under varying values of λ, highlighting the significance of SS in influencing BCD outcomes. On WHU-CD, the model achieves peak performance when λ is set to 0.5 for the BCD task. This indicates that assigning a moderate weight to the segmentation loss is optimal for this dataset. The model performs best on the more challenging SECOND and HRSCD datasets with unbalanced classes and more semantic categories when λ takes smaller values of 0.01 or 0.1. The SS predictions are less reliable on SECOND compared to WHU-CD. Therefore, the results presented in Table XIII also underscore that if the performance of the SS task falls short within the multitask framework, its ability to augment the BCD task's performance is compromised.

4) Impact of the Segmentation Loss Type: We conducted a comparative study to evaluate the impact of different loss functions. Specifically, we replaced the standard CE loss with weighted CE (WCE) [45], [46] for the SS task by incorporating class ratio priors computed from the training data. The WCE loss allows the model to assign higher importance to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

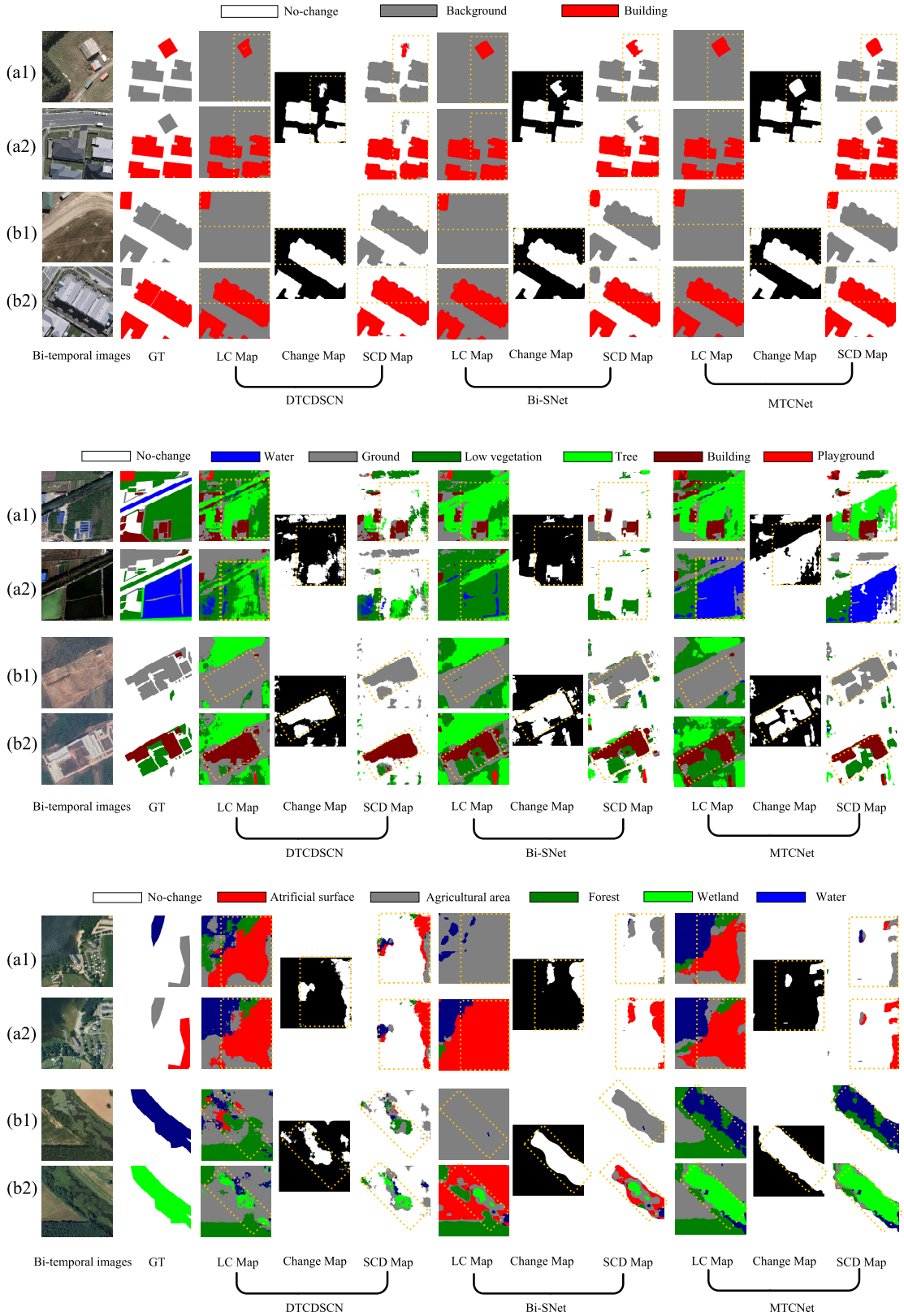


Fig. 11. Visual comparison of SCD results. Three sets of (a1) and (a2), and (b1) and (b2) (top to bottom) correspond to the comparison results on the WHU-CD, SECOND, and HRSCD datasets. GT for SCD was obtained by comparing the annotated land-cover categories.

rare classes, thereby improving its ability to classify semantic changes accurately.

Tables XIV–XVII present the BCD and SS results using different loss functions. Table XIV shows that using the WCE loss did not substantially improve the model's BCD performance. However, Table XV indicates that the WCE loss can somewhat improve the SS performance on the SECOND and HRSCD datasets, which exhibit imbalanced class distributions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

TABLE XV
SS RESULTS USING DIFFERENT LOSSES (%)

Dataset	Loss	OA	F1	mIoU	Kappa
WHU-CD	CE	98.66	97.68	95.50	95.35
	WCE	**98.79**	**97.81**	**95.76**	**95.63**
SECOND	CE	77.65	69.87	54.84	68.78
	WCE	**78.10**	**71.26**	**56.38**	**69.53**
HRSCD	CE	87.07	75.52	62.68	76.19
	WCE	**87.38**	**77.32**	**64.64**	**76.62**

TABLE XVI
CLASS-WISE SS RESULTS USING DIFFERENT LOSSES ON SECOND (%)

Metric	Loss	Class					
		water	Ground	Low vegetation	Tree	Building	playground
Precision	CE	**64.70**	80.16	68.23	**62.96**	**87.04**	**75.99**
	WCE	57.16	**81.33**	**70.81**	60.96	86.18	67.69
Recall	CE	46.00	78.35	**70.88**	62.12	88.51	58.93
	WCE	**55.16**	**78.64**	68.32	**67.14**	**89.66**	**72.75**
F1	CE	53.77	79.24	69.53	62.54	87.77	66.38
	WCE	**56.14**	**79.96**	**69.54**	**63.9**	**87.89**	**70.13**
IoU	CE	36.77	65.62	53.29	45.49	78.20	49.68
	WCE	**39.03**	**66.62**	**53.30**	**46.95**	**78.39**	**54.00**

TABLE XVII
CLASS-WISE SS RESULTS USING DIFFERENT LOSSES ON HRSCD (%)

Metric	Loss	Class				
		Artificial surface	Agricultural area	Forest	Wetland	Water
Precision	CE	**86.91**	89.40	63.73	44.86	**82.91**
	WCE	85.92	**89.44**	**75.29**	**52.89**	82.36
Recall	CE	83.93	89.86	**81.70** 5 / 16	**54.19**	80.96
	WCE	**84.87**	**90.05**	76.66	53.90	**81.86**
F1	CE	85.39	89.63	71.60	49.08	81.92
	WCE	**85.39**	**89.75**	**75.97**	**53.39**	**82.11**
IoU	CE	74.51	81.21	55.77	32.52	69.38
	WCE	**74.51**	**81.40**	**61.25**	**36.42**	**69.65**

TABLE XVIII
IMPACT OF THE NUMBER OF FEATURE SCALES ON SECOND

Model	Branch				BCD		SS	
	Res2	*Res3*	*Res4*	*Res5*	F1	IoU	mIoU	Kappa
MCTNet	×	×	×	✓	67.49	50.93	52.51	65.92
MCTNet	×	✓	×	✓	69.84	53.65	53.42	67.72
MCTNet	✓	×	×	✓	69.49	53.25	53.05	67.02
MCTNet	✓	✓	×	✓	**70.37**	**54.28**	**56.38**	**69.53**
MCTNet	✓	✓	✓	✓	67.61	51.07	53.84	66.97

SS is the abbreviation for semantic segmentation.

TABLE XIX
IMPACT OF THE NUMBER OF FEATURE SCALES ON HRSCD

Model	Branch				BCD		SS	
	Res2	*Res3*	*Res4*	*Res5*	F1	IoU	mIoU	Kappa
MCTNet	×	×	×	✓	74.12	58.88	61.93	75.27
MCTNet	×	✓	×	✓	74.22	59.01	60.17	74.94
MCTNet	✓	×	×	✓	74.25	59.04	61.29	75.39
MCTNet	✓	✓	×	✓	**74.38**	**59.22**	**64.64**	**76.62**
MCTNet	✓	✓	✓	✓	72.20	56.49	60.21	72.74

TABLE XX
ABLATION RESULTS FOR THE BCD TASK (%)

TF	MF	SFL	ML	WHU-CD		SECOND		HRSCD	
				F1	IoU	F1	IoU	F1	IoU
×	✓	✓	✓	93.28	87.41	69.68	53.47	74.11	58.87
✓	×	✓	✓	92.83	86.61	69.66	53.45	72.48	56.83
✓	✓	×	✓	92.96	86.84	69.21	52.92	73.64	58.28
✓	✓	✓	×	91.78	84.81	69.94	53.78	74.09	58.84
✓	✓	✓	✓	**94.35**	**89.30**	**70.37**	**54.28**	**74.38**	**59.22**

TF stands for Transformer. MF Stands for multiscale feature fusion. ML stands for multitask learning.

On the WHU-CD dataset, the WCE loss slightly affects the SS task, probably because there are only two categories of *building* and *nonbuilding* on this dataset, and the performance of the SS task is saturated using the CE loss. As shown in Tables XVI and XVII, using the WCE loss significantly for the SS task improves the recall of rare classes like *water* and *playground* on the SECOND dataset. On the HRSCD dataset, the $F1$ and IOU metrics are improved, particularly for the *forest* and *wetland* classes. These results highlight the WCE loss's effectiveness in improving SS performance on imbalanced datasets.

5) Impact of the Number of the Feature Scales: We extracted multiscale feature maps from the four residual blocks with varying scales to evaluate the impact of the number of feature scales extracted from the backbone network. Blocks *Res4* and *Res5* yield the same output feature scale. For the output feature of *Res4*, we integrate it into the features extracted from other residual blocks during the multiscale feature fusion stage. As illustrated in Tables XVIII and XIX, the model achieves peak performance when utilizing three feature scales, extracted from residual blocks *Res2*, *Res3*, and *Res5*. With fewer scales, the model may struggle to adequately represent features at different levels of granularity, potentially compromising its ability to detect subtle changes in RS imagery. Extracting features from *Res4* and fusing them with the output features of other residual modules does not improve the model's performance but diminishes it. The decline in performance when using feature maps extracted from all four residual modules could be attributed to several factors. First, the additional feature maps may introduce redundancy or noise, diluting the discriminative information required for effective CD. Second, the integration process of these feature maps may not be optimal, leading to interference or conflicting signals among the different scales.

E. Ablation Studies

We conducted ablation studies to evaluate the impact of four key components in MCTNet: the Transformer module, multiscale feature fusion, SFL, and MTL. We used CE loss on WHU-CD and WCE loss on SECOND and HRSCD for the SS task.

To assess the significance of each module, we conducted a series of experiments where we removed individual components from our network architecture and evaluated its performance. Specifically, we tested the network with this module removed to evaluate the Transformer module. Similarly, to assess the importance of the multiscale feature fusion strategy, we eliminated it from $F_i^{j(2)}$ to $F_i^{j(3)}$. For the SFL module, we removed it and directly concatenated the feature maps $F_1^{j(4)}$ to $F_2^{j(4)}$ along the channel dimension. Following this, we ensured that both the BCD and SS heads maintained the same structure, except for the dimensions of the input and output channels. Finally, to gauge the effectiveness of MTL, we retained only one type of task head in the output layer.

Table XX presents the ablation results for the BCD task. On WHU-CD, the Transformer module, the multiscale mechanism, the SFL module, and MTL improve the $F1$ score by 1.07%, 1.52%, 1.39%, and 2.57%, respectively. They also improve IoU by 1.89%, 2.69%, 2.46%, and 4.49%, respectively. These four key components all improved the IoU and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

TABLE XXI
ABLATION RESULTS FOR THE SS TASK (%)

TF	MF	SFL	ML	WHU-CD		SECOND		HRSCD	
				mIou	Kappa	mIoU	Kappa	mIoU	Kappa
×	✓	✓	✓	95.43	95.27	55.60	68.32	62.36	76.61
✓	×	✓	✓	94.97	94.78	54.72	67.24	60.04	75.18
✓	✓	×	✓	95.19	95.03	56.28	68.85	62.13	75.42
✓	✓	✓	×	93.29	92.54	56.30	68.39	63.60	75.39
✓	✓	✓	✓	**95.50**	**95.35**	**56.38**	**69.53**	**64.64**	**76.63**

$F1$ scores to some extent on SECOND and HRSCD. However, MTL had a lesser effect on SECOND and HRSCD, likely due to lower SS performance. Table XXI depicts the ablation results for the SS task. Across all configurations on WHU-CD, high performance is observed, and the removal of any component had minimal impact due to saturation. A decrease in performance was noted after removing the multiscale feature fusion, which enhances the model's ability to understand and process images at different scales.

V. CONCLUSION

This article introduces MCTNet, a MCTNet for SCD using bitemporal RS images. MCTNet combines SS and BCD tasks in a triple-branch framework. Extensive experiments on the WHU-CD, SECOND, and HRSCD datasets show MCTNet's effectiveness and efficiency in SCD. Ablation studies highlight the importance of the Transformer module, multiscale feature fusion, SFL, and MTL in boosting model performance.

We recognize that using a ResNet backbone pretrained on natural images may limit performance due to its inadequate capture of RS data characteristics. Future work will explore using foundation models like the segment anything model [47], trained on RS images, as the backbone. We will also apply semi-supervised domain adaptation techniques to utilize labeled source data, limited labeled target data, and abundant unlabeled target data to boost performance on the target domain. These improvements aim to enhance the effectiveness and robustness of SCD methods in RS applications.

REFERENCES

[1] L. Bruzzone, D. F. Prieto, and S. B. Serpico, "A neural-statistical approach to multitemporal and multisource remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1350–1359, May 1999.

[2] L. Bruzzone, R. Cossu, and G. Vernazza, "Detection of land-cover transitions by combining multidate classifiers," *Pattern Recognit. Lett.*, vol. 25, no. 13, pp. 1491–1500, Oct. 2004.

[3] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[4] Y. Niu, H. Guo, J. Lu, L. Ding, and D. Yu, "SMNet: Symmetric multi-task network for semantic change detection in remote sensing images based on CNN and transformer," *Remote Sens.*, vol. 15, no. 4, p. 949, Feb. 2023.

[5] W. Liu, Y. Lin, W. Liu, Y. Yu, and J. Li, "An attention-based multiscale transformer network for remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 599–609, Aug. 2023.

[6] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[7] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[8] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multiscale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.

[9] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[10] H. Li, X. Liu, H. Li, Z. Dong, and X. Xiao, "MDFENet: A multiscale difference feature enhancement network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3104–3115, 2023.

[11] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Sep. 2019.

[12] Y. Yuan and L. Lin, "Self-supervised pretraining of transformers for satellite image time series classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 474–487, 2021.

[13] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, Feb. 2021.

[14] K. Xu, P. Deng, and H. Huang, "Vision transformer: An excellent teacher for guiding small networks in remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618715.

[15] Z. Li, G. Chen, and T. Zhang, "A CNN-transformer hybrid approach for crop classification using multitemporal multisensor images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 847–858, 2020.

[16] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[17] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[18] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[19] X. Song, Z. Hu, and J. Li, "PSTNet: Progressive sampling transformer network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8442–8455, 2022.

[20] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," 2022, *arXiv:2209.08290*.

[21] A. Singh, "Digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jun. 1989.

[22] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[23] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.

[24] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[25] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[26] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102465.

[27] H. Xia, Y. Tian, L. Zhang, and S. Li, "A deep Siamese postclassification fusion network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622716.

[28] S. Tian, X. Tan, A. Ma, Z. Zheng, L. Zhang, and Y. Zhong, "Temporal-agnostic change region proposal for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 306–320, Oct. 2023.

[29] Q. Shen, J. Huang, M. Wang, S. Tao, R. Yang, and X. Zhang, "Semantic feature-constrained multitask Siamese network for building change detection in high-spatial-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 78–94, Jul. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.

[30] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.
[31] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.
[32] F. Cui and J. Jiang, "MTSCD-net: A network based on multi-task learning for semantic change detection of bitemporal remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103294.
[33] Z. Li, X. Wang, S. Fang, J. Zhao, S. Yang, and W. Li, "A decoder-focused multitask network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609115.
[34] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.
[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[36] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," 2016, *arXiv:1606.08415*.
[37] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.
[38] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.
[39] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.
[40] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
[41] J. Xu, C. Luo, X. Chen, S. Wei, and Y. Luo, "Remote sensing change detection based on multidirectional adaptive feature fusion and perceptual similarity," *Remote Sens.*, vol. 13, no. 15, pp. 3053–3056, 2021.
[42] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.
[43] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.
[44] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.
[45] W. Liu, Z. Luo, and S. Li, "Improving deep ensemble vehicle classification by using selected adversarial samples," *Knowl.-Based Syst.*, vol. 160, pp. 167–175, Nov. 2018.
[46] W. Liu et al., "Adversarial unsupervised domain adaptation for 3D semantic segmentation with multi-modal learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 211–221, Jun. 2021.
[47] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.



Wei Liu (Member, IEEE) received the Ph.D. degree in artificial intelligence from Xiamen University, Xiamen, China, in 2015.

Currently, he is an Associate Professor with East China Jiaotong University, Nanchang, China. He has authored or co-authored top remote sensing journals, including *International Journal of Applied Earth Observation and Geoinformation* (JAG) and *ISPRS Journal of Photogrammetry and Remote Sensing* (ISPRS JPRS). He serves as the Guest Editor of JAG. His research focuses on remote sensing image processing, machine learning, and computer vision.

Ziwen Kang, photograph and biography not available at the time of publication.



Jiawei Liu received the master's degree from East China Jiaotong University, Nanchang, China, in 2023.

Currently, he is an Assistant Teacher with Pingxiang University, Pingxiang, China. His research focuses on remote sensing image processing, machine learning, and computer vision.

Yiyuan Lin, photograph and biography not available at the time of publication.



Yongtao Yu (Senior Member, IEEE) received the Ph.D. degree in computer science and technology from Xiamen University, Xiamen, China, in 2015.

He is currently an Associate Professor with the Faculty of Computer and Software Engineering, Huaiyin Institute of Technology, Huaian, China. He has authored or co-authored more than 60 research articles published in refereed journals. His research interests include intelligent transportation systems, computer vision, deep learning, and intelligent interpretation of 3-D point clouds and remotely sensed imagery.

Dr. Yu serves as an Editorial Board Member for *International Journal of Applied Earth Observation and Geoinformation* (JAG) and an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (IEEE TGRS), and IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS (IEEE TITS).



Jonathan Li (Fellow, IEEE) received the Ph.D. degree in geomatics engineering from the University of Cape Town, Cape Town, South Africa, in 2000.

He is currently a Professor of geomatics and systems design engineering with the University of Waterloo, Waterloo, ON, Canada. He has co-authored more than 650 publications, more than 250 of which were published in top remote sensing journals, including *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing* (ISPRS JPRS), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (IEEE TGRS), and *International Journal of Applied Earth Observation and Geoinformation* (JAG). He has also published papers in flagship conferences in computer vision and AI, including Conference on Computer Vision and Pattern Recognition (CVPR), NeurIPS, Association for the Advancement of Artificial Intelligence (AAAI), and International Joint Conference on Artificial Intelligence (IJCAI). He has supervised nearly 200 master's/Ph.D. Students as well as a Postdoctoral Fellows/Visiting Scholars to completion. His main research interests include AI-based information extraction from Earth observation images and LiDAR point clouds, pointgrammetry and remote sensing, GeoAI and 3-D vision for digital twin cities and autonomous driving.

Dr. Li is a fellow of the Canadian Academy of Engineering, the Royal Society of Canada (Academy of Science), and the Engineering Institute of Canada. He is the President of the Canadian Institute of Geomatics (CIG). He is the Editor-in-Chief for JAG and an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (IEEE TGRS), and IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS (IEEE TITS).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:22:41 UTC from IEEE Xplore. Restrictions apply.